

Notes: Degryse & Ongena (JF, 2005)

Distance, Lending Relationships, and Competition

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1 Big picture

- **Question.** How does physical distance between a borrowing firm, its lending bank branch, and competing bank branches affect loan interest rates?
- **Core hypothesis.** If banks can price discriminate spatially, close borrowers may pay higher rates because they face higher total costs of switching to more distant competitors. Conversely, borrowers closer to competitors should receive lower rates because the lending bank faces more competitive pressure.
- **Main empirical object.** The paper studies more than 15,000 Belgian small-business loan contracts from the complete loan portfolio of a large bank. The key dependent variable is the loan rate until the next revision, measured in basis points.
- **Key distance variables.** The first distance is shortest travel time from the borrower to the lending branch. The second is shortest travel time from the borrower to the closest quartile competitor in the borrower's postal zone.
- **Main result.** Loan rates decrease with distance to the lender and increase with distance to competitors. In the main specifications, the coefficients on the two distance variables are roughly equal in absolute value and opposite in sign.
- **Interpretation.** The sign pattern fits spatial price discrimination based on borrower transportation costs. It is hard to reconcile with a simple monitoring-cost story, which would predict higher rates for more distant borrowers.
- **Role of relationship lending.** Main-bank status lowers loan rates, but relationship duration raises rates. The distance pattern is weaker for broader or more collateralized relationships, suggesting that location rents and relationship rents are distinct.
- **Competition.** The number of local competitors has little effect, while branch concentration has a small positive effect. The most informative competition variable is not simply the number of branches but the travel-time distance to alternative lenders.
- **Mechanism.** Spatial price discrimination is strongest for transactional, short-term, uncollateralized, term, and noncapital-expenditure loans. It is weaker for lines of credit, collateralized loans, and loans supporting capital expenditures.
- **Economic magnitude.** The authors interpret the distance coefficients as plausible transportation-cost based rents. Their estimates imply location rents of around four percent of the bank's marginal cost of funding.
- **Takeaway.** Even in a densely branched Belgian banking market, geographic proximity remains economically meaningful. The paper cautions against presaging the "death of distance" in European banking.

2 Incremental contribution of the paper

2.1 Contribution to bank competition and relationship lending

The paper's incremental contribution is to separate two sources of bank market power that are often mixed together in the banking literature:

- **relationship or information rents**, which arise because the bank learns about the borrower over time;
- **location rents**, which arise because the borrower is physically closer to one bank branch than to alternative branches.

Interpretation: Prior relationship-lending studies often find that banks extract rents from informational lock-in. Degryse and Ongena show that a bank can also extract rents before or apart from informational lock-in, simply because the borrower is locally captive.

2.2 Contribution to empirical price discrimination

The paper contributes to the broader price-discrimination literature by studying a continuous and economically interpretable discriminatory characteristic: travel time. Unlike race, gender, or other social categories, distance can be measured continuously and can be mapped directly into a primitive cost of switching or visiting branches.

Interpretation: This makes the evidence unusually interpretable. A coefficient on distance can be translated into a transportation-cost story, while a coefficient on a social category usually cannot be benchmarked as cleanly.

2.3 Contribution to geography and finance

The paper contributes to the geography-of-finance literature by showing that distance affects not only lending availability but also contract pricing. It complements work on geographical distance in bank lending, venture capital, mutual funds, cross-border banking, and investor home bias.

Interpretation: The point is not merely that borrowers choose nearby banks. The paper shows that, conditional on many contract and borrower characteristics, nearby alternatives affect the price of credit.

2.4 Contribution to measuring local competition

A key novelty is that the paper does not rely only on the number of competitors or HHI. It directly measures how far the borrower is from the lending branch and from competitor branches. This reveals that the *spatial location* of competitors is more informative than their count.

Interpretation: Two postal zones can have the same number of bank branches, but very different competitive pressure if competitor branches are far from the borrower. The paper therefore moves from market-structure counts to customer-level competitive geography.

2.5 Contribution to interpreting branch networks

The results imply that brick-and-mortar branches remain economically relevant for small firms. Branch networks may reduce borrowing costs by placing alternative lenders physically near borrowers, thereby weakening spatial price discrimination.

Interpretation: This is a policy and strategy contribution. Branch closures and bank consolidation may raise the value of local market power even if credit scoring and communications technology improve.

3 Data and measurement

3.1 Institutional setting

The empirical setting is Belgium in the 1990s. The country is small, densely populated, and heavily bank-based. The analyzed bank has many branches and a lending technology that gives local loan officers discretion in evaluating and pricing loan applications.

Interpretation: This setting is useful because distances are small but still measurable. If distance matters even in Belgium, then location may matter in many other banking markets as well.

3.2 Loan contract sample

The final sample contains 15,044 loan contracts after excluding contracts with mapping problems, address problems, outlying travel distances, relocated borrowers, and contracts in postal zones without identified competing bank branches. The loans are primarily to small firms; about 83 percent of firms in the portfolio are single-person businesses.

- The dependent variable is the loan rate until the next revision, measured in basis points.
- Loan size averages about 0.88 million Belgian francs, with a median of 0.30 million.
- Repayment duration averages 2.35 years, with a median of 0.55 years.
- Collateral is present in about 26 percent of loan contracts.

Interpretation: The sample is well suited for studying small-business relationship lending because it contains detailed contract-level characteristics and local borrower-bank geography.

3.3 Geographical distance measures

For each borrower, the authors calculate shortest travel time by car:

- from the borrower to the lending bank branch;
- from the borrower to competitor bank branches in the same postal zone.

The main competitor-distance variable is the distance to the *closest quartile competitor*, namely the 25th percentile travel time among competitor branches in the borrower’s postal zone.

Interpretation: The closest-quartile measure is less sensitive to mapping errors, address errors, or one branch that is technically close but not a true product competitor. It is a more stable proxy for local competitive pressure.

3.4 Relationship variables

The paper uses two relationship variables:

- **Main bank:** equals one if the bank considers itself the borrower’s main bank.
- **Duration of relationship:** the number of years the borrower has had a relationship with the current lender.

Interpretation: Main-bank status captures breadth and importance of the relationship. Relationship duration captures time-based lock-in and accumulated private information.

3.5 Competition variables

The competition measures are:

- number of competing bank branches in the borrower’s postal zone;
- a Herfindahl-Hirschman Index based on bank branch shares within the postal zone;
- postal-zone random effects in some specifications;
- average real-estate price and urban dummies as local location controls.

Interpretation: Counting branches and measuring branch concentration provide market-structure controls, but the paper’s central argument is that distance to the actual alternatives is the relevant competitive pressure.

3.6 Firm, loan, and local controls

The specifications control for loan purpose, collateral, repayment duration, loan revisability, loan size, firm type, postal area, industry, interest-rate benchmarks, term spreads, year dummies, and local characteristics.

Interpretation: These controls reduce the possibility that distance simply proxies for borrower type, loan type, or the local economic environment.

4 Models and empirical specifications

4.1 Baseline loan-pricing equation

A compact version of the main empirical specification is:

$$LoanRate_{ijz} = \alpha + \beta_L \log(1 + DistanceToLender_{ijz}) + \beta_C \log(1 + DistanceToCompetitors_{iz}) + \Gamma' X_{ijz} + \varepsilon_{ijz}, \quad (1)$$

where i indexes the borrower, j the lending branch, and z the postal zone. X_{ijz} contains relationship, loan, firm, interest-rate, local-market, industry, and postal-area controls.

Interpretation: The two key coefficients are β_L and β_C . Spatial price discrimination predicts $\beta_L < 0$ and $\beta_C > 0$.

4.2 Spatial price discrimination hypothesis

Under borrower transportation costs and spatial price discrimination:

$$\beta_L < 0, \quad \beta_C > 0. \quad (2)$$

- A borrower close to the lending bank has less incentive to visit other banks, so the lender can charge a higher rate.
- A borrower close to competing banks has better outside options, so the lender must charge a lower rate.

Interpretation: The opposite signs are central. A negative coefficient on distance to lender alone could be interpreted many ways. The simultaneous positive coefficient on distance to competitors gives the result a spatial-competition interpretation.

4.3 Monitoring-cost alternative

A simple monitoring-cost model predicts:

$$\beta_L > 0, \quad (3)$$

if monitoring costs rise with distance from the borrower to the lending branch. Some models with bank monitoring costs and discriminatory pricing can also imply $\beta_L < 0$ and $\beta_C > 0$, but the paper uses heterogeneity across loan types to assess whether monitoring or transportation costs are more plausible.

Interpretation: If information deteriorates with distance, far borrowers should often look riskier or more costly. The empirical finding that more distant borrowers often pay lower rates is difficult for a simple monitoring-cost view.

4.4 Relationship and interaction specifications

The paper adds relationship interactions:

$$LoanRate = \dots + \delta_1 MainBank + \delta_2 Duration + \phi_1 MainBank \times \log(1 + DistanceToLender) + \phi_2 MainBank \times \log(1 + DistanceToCompetitors) \quad (4)$$

Interpretation: These interactions ask whether spatial discrimination is softened when the borrower has a main-bank relationship. The paper finds that main-bank interactions partially offset the distance effects.

4.5 Competition specifications

The paper compares:

- number of competitors;
- HHI based on branch shares;
- postal-zone random effects;
- real-estate and urban controls.

Interpretation: The number of competitors is often not significant. HHI is small but positive. Distance to competitors remains the economically central competition variable.

4.6 Subsample specifications

The paper estimates variants by:

- loan size;
- loan duration;
- collateral status;
- loan type: lines of credit versus term loans;
- capital versus noncapital expenditure purpose;
- firm type;
- relative distance to lender and competitors.

Interpretation: The subsample analysis is the main mechanism test. Transportation costs predict stronger discrimination when switching or visiting competitors is more relevant and when borrower-bank relationships are thinner.

5 Identification and empirical design

5.1 Core identification idea

The identification strategy is reduced-form and cross-sectional. The authors compare loan rates across otherwise similar borrowers and loan contracts that differ in travel time to the lender and travel time to competing bank branches.

Interpretation: The paper does not claim random assignment of distance. Instead, it relies on a rich contract-level data set, detailed geography, and mechanism tests that separate transportation-cost and information interpretations.

5.2 Why using both distance measures matters

The key empirical design feature is that the authors include both borrower-lender distance and borrower-competitor distance. This permits a direct test of spatial price discrimination: rates should fall with distance to the lender and rise with distance to competitors.

Interpretation: If the lending bank charges close borrowers more because they are locally captive, both coefficients should move in opposite directions. This pairwise sign pattern is much more informative than either distance coefficient alone.

5.3 Addressing competition and local-market confounds

The paper includes postal-area and industry dummies, interest-rate controls, loan characteristics, firm characteristics, HHI, number of competitors, postal-zone random effects, average real-estate prices, and urban indicators.

Interpretation: These controls are meant to distinguish spatial price discrimination from local demand, local property costs, borrower sector, loan risk, and broad market concentration.

5.4 Mechanism tests

The most important mechanism tests are the subsample splits. The paper finds stronger spatial price discrimination for transactional, short-term, uncollateralized, term, and noncapital expenditure loans. Discrimination relaxes for broader relationships, collateralized loans, lines of credit, and capital expenditure loans.

Interpretation: These patterns line up with transportation-cost based price discrimination because the effect is strongest when switching to alternative banks is a salient fixed cost. If the main force were deteriorating information with distance, the same heterogeneity would be harder to explain.

5.5 Robustness checks

The paper checks:

- travel time in logs and levels;
- actual closest competitor instead of closest-quartile competitor;
- fastest driving distance in kilometers instead of travel time;

- contracting-year and branch-network changes;
- postal zones with no competing branch;
- additional firm characteristics and accounting measures.

Interpretation: The robustness checks are important because distance could otherwise proxy for technology, borrower selection, omitted borrower quality, or branch location choices.

5.6 Limitations

- Distances are not randomly assigned.
- Borrower branch choice may be endogenous.
- Some borrower characteristics are unobservable to the econometrician but observable to the bank.
- The data come from one large Belgian bank, so external validity to other countries and modern digital banking markets is not automatic.
- The mechanism is inferred from sign patterns and heterogeneity, not from direct observation of borrower search costs.

Interpretation: The paper is strongest as evidence of a coherent spatial pricing pattern in a uniquely detailed setting, not as a randomized experiment.

6 Section-by-section study guide

6.1 Introduction

The introduction frames the paper around market power in credit markets. It emphasizes that banks can obtain rents either from physical proximity or from private information. The central contribution is showing that location rents are empirically important and distinct from relationship rents.

6.2 Section I: Theoretical predictions

This section reviews theories that link loan rates and distance. The main takeaway is that different theories imply different signs for distance to the lender, distance to competitors, and the number of competitors. Table I is the map that connects empirical coefficients to mechanisms.

6.3 Section II: Data and methodology

This section introduces the loan-level data, bank-branch geography, borrower postal zones, and travel-time calculations. It also explains why the closest-quartile competitor is used and why the final sample focuses on postal zones with competitors.

6.4 Section III: Empirical results

This section reports the main regressions. First, the authors document relationship, loan, and firm controls. Then they add competition and distance. Finally, they test heterogeneity by relationship type, loan type, collateral, duration, loan purpose, and firm characteristics.

6.5 Section IV: Conclusion

The conclusion argues that spatial price discrimination, not a simple information-cost mechanism, best explains the empirical patterns. It also emphasizes that branch networks can remain relevant even when communication technology improves.

7 Tables and figures

7.1 Table I: Theoretical Models Linking Loan Rates and Distance + Table II: Bank and Postal Zone Statistics

Read:

- Table I lists theoretical predictions for how loan rates should depend on distance to the lender, distance to competitors, and the number of competitors.
- Spatial price discrimination predicts that loan rates fall with distance to the lender and rise with distance to the closest competitors.
- Monitoring-cost marginal pricing predicts the opposite sign for lender distance: if monitoring costs rise with borrower distance, loan rates should rise with lender distance.
- Table II reports 145 banks and 7,477 branches in Belgium. It also reports 837 postal zones with bank branches and 921 postal zones with borrowers of the bank.
- The average postal zone has 6.4 banks; adjacent postal zones with bank branches contain on average 53.6 banks.

Detailed interpretation: Table I is the theoretical compass for the paper. The main empirical finding, $\beta_L < 0$ and $\beta_C > 0$, lines up with discriminatory pricing rather than simple uniform pricing or simple monitoring-cost pass-through. Table II establishes that the empirical setting has enough geographical variation and enough branch density to identify local competitive pressure.

Economic message: Spatial competition is not simply a question of how many banks exist nationally. The relevant object is where branches are located relative to each borrower.

Table I Theoretical Models Linking Loan Rates and Distance			
The table lists models, categorized by argumentation, hypothesizing the impact of distance for a given number of competitors on the loan rate.			
Arguments & Discussed Models	Impact on the Loan Rate of the		
	Distance to the Lender	Distance to the Closest Competitor	Number of Competitors
<i>Transportation Costs (for borrower)</i>			
Uniform pricing	no	no	negative
Discriminatory pricing	negative	positive	negative
<i>Monitoring Costs (for lender)</i>			
Marginal cost pricing	positive	negative	negative
Discriminatory pricing	negative	positive	negative
	Distance to the Relationship Bank	Distance to the Transactional Bank	Number of Competitors
<i>Asymmetric Information</i>			
Dell'Ariccia (2001)	negative	no	negative
Hauswald and Marquez (2003)	negative	positive	positive/negative

(a) Table I: theoretical predictions

Table II Bank and Postal Zone Statistics					
The table provides key statistics about the lending bank and the Belgian postal zones/areas.					
Total number of banks	145				
Total number of bank branches	7,477				
	Postal Zones		Postal Areas		
Total number with bank branches	837		9		
Total number with borrowers of the bank	921		9		
Total number	1,168		9		
Average surface area, in km ²	26		3,359		
Average population	8,632		1,120,209		
	Mean	Median	Minimum	Maximum	SD
Number of banks per postal zone	6.4	4	0	103	10.4
Number of adjacent postal zones / postal zone with bank branches	5.1	5	0	16	2.0
Number of banks in postal zones adjacent to postal zones with bank branches	53.6	44	2	471	42.4

(b) Table II: bank and postal-zone statistics

7.2 Table III: Data Description

Read:

- The final estimating sample contains 15,044 loan contracts.
- Median distance to lender is 4.29 minutes; mean distance is 6.90 minutes.
- Median distance to closest competitors is 3.27 minutes; mean distance is 3.82 minutes.
- The bank is the main bank in 58.82 percent of observations.
- Average relationship duration is 7.93 years.
- Average number of competitors in the borrower's postal zone is 17.18.
- Mean loan rate is 812 basis points, with median 782 basis points.
- Most borrowers are small firms; small-firm dummy mean is 15.99 percent under the bank's size definition.

Detailed interpretation: Table III shows that borrowers are close to both lenders and competitors, which makes the economic significance of distance striking: small travel-time differences are enough to affect loan rates. It also shows that relationship and loan contract characteristics vary substantially across loans, justifying detailed controls.

Economic message: The empirical setting is not one of remote rural lending. It is a dense banking market where even a few minutes of travel time can change the distribution of bargaining power.

Table III
Data Description

The table defines the variables employed in the empirical specifications and provides their mean, median, minimum, maximum, and standard deviation. The number of observations is 15,044.

Variables	Definition	Mean	Median	Minimum	Maximum	SD
Geographical Distance						
Distance to lender	Shortest traveling time, in minutes	6.90	4.29	0.00	51.90	7.30
Distance to closest competitors	Shortest traveling time to the closest quartile competitor in the borrower's postal zone, in minutes	3.82	3.27	0.00	24.00	2.33
Relationship Characteristics						
Main bank	= 1 if bank considers itself as main bank, ^a in %	58.82	100	0	100	49.22
Duration of relationship	Length of relationship with current lender, in years	7.93	7.47	0.00	26.39	5.44
Competition Measures						
Number of competitors	Number of branches (minus the lender's) in the borrower's postal zone	17.18	13	1	103	15.49
Herfindahl-Hirschman Index	Summed squares of bank market shares, by number of branches, in each postal zone	0.17	0.15	0.05	1.00	0.11
Loan Rate and Size						
Loan rate	Interest rate on loan until next revision, in basis points	812	782	200	2,200	236
Loan size	Size of loan, in millions of BEF ^b	0.88	0.30	0.005	80	1.83
Loan Contract Characteristics						
Including Four Loan Revisability Dummies	= 1 if loan is a business term loan (investment credit)					
Collateral	= 1 if loan is secured via collateral, in %	26.40	0	0	100	44.08
Repayment duration of loan	Repayment duration of loan, in years	2.35	0.55	0.00	20.00	3.26
Loan Purpose						
Mortgage	= 1 if loan is a business mortgage loan	<i>n/a</i> ^c				
Term	= 1 if loan is a business term loan (investment credit)	<i>n/a</i> ^c				
Securitizable term	= 1 if loan is a securitizable business term loan (investment credit)	<i>n/a</i> ^c				
Bridge	= 1 if loan is a bridge loan	<i>n/a</i> ^c				
Prepay taxes	= 1 if loan is credit to prepay taxes	<i>n/a</i> ^c				
Consumer credit	= 1 if loan is a consumer credit loan (capturing installment loans)	<i>n/a</i> ^c				
Other	= 1 if loan is given for another purpose or its purpose is not specified	<i>n/a</i> ^c				
Rollover	= 1 if loan is given to prepay another loan, in %	10.20	0	0	100	30.27

Firm Characteristics	Including 8 Postal Area and 49 Industry Dummies					
Small firm	= 1 if < 10 employees and turnover < 250 million BEF ^b in %	15.99	0	0	100	36.64
Medium firm	= 1 if > 10 employees or turnover > 250 million BEF ^b in %	0.89	0	0	100	9.40
Large firm	= 1 if turnover > 1 billion BEF ^b in %	0.14	0	0	100	3.73
Limited partnership	= 1 if firm is limited partnership, in %	11.97	0	0	100	32.46
Limited partnership w/ES	= 1 if firm is limited partnership with equal sharing, in %	1.18	0	0	100	10.78
Corporation	= 1 if firm is corporation, in %	3.78	0	0	100	19.07
Temporary arrangement	= 1 if firm is a temporary arrangement, in %	0.85	0	0	100	9.18
Firm Location						
Average real estate price	In the Postal Zone in 1996, in millions of BEF ^b	2.40	2.19	0.35	7.84	0.99
Urban	= 1 if located in agglomeration with > 250,000 inhabitants, in %	9.73	0	0	100	29.64
Interest Rate Variables	Including 2-Year Dummies					
Government security	Interest rate on a Belgian government security with equal repayment duration as loan to firm, in basis points	389	350	305	805	87
Term spread	Yield on Belgian government bond of 5 years—yield on Treasury bill with maturity of 3 months, in basis points	179	177	100	268	31

^aThe definition used by the bank to determine whether it is the main bank is: for single-person businesses and small firms, these have a turnover on the current account of at least BEF 100,000 per month and buy at least two products from that bank.
^bFor bank-strategic considerations we cannot reveal the relative importance of the types of loans.
^cForty Belgian Francs (BEF) are approximately equal to US\$1 during the sample period.

Table III: Data Description, part 2

Table III: Data Description, part 1

7.3 Table IV: Borrowing Costs, Firm, and Loan Characteristics

Read:

- Main-bank borrowers pay about 40.7 basis points less, suggesting that broad relationships are valuable to borrowers.
- Relationship duration raises loan rates; the coefficient on log relationship duration is 19.2 basis points.
- Collateralized loans have lower rates, with a coefficient around -50.9 basis points.
- Longer repayment duration is associated with lower rates.
- Larger firms pay lower rates; large-firm coefficient is about -170 basis points.
- Loan purpose, postal area, industry, year, and interest-rate variables are controlled for.

Detailed interpretation: Table IV establishes the baseline pricing environment before the distance variables are introduced. The controls have intuitive signs: safer or more established borrowers and secured loans have lower rates. The relationship-duration coefficient is positive, which is consistent with relationship lock-in or the extraction of information rents over time.

Table IV
Borrowing Costs, Firm, and Loan Characteristics

The table lists the coefficients from a regression with the *loan rate* until the next revision, in basis points, as the dependent variable. *Main bank* equals 1 if the bank considers itself as the main bank and 0 otherwise. *Duration of relationship* is the length of relationship with the current lender, in years. *Collateral* equals 1 if the loan is secured via collateral and 0 otherwise. *Repayment duration of loan* is in years. The *loan purpose* and *firm characteristics* variables are all dummies that equal 1 if the loan or firm has the featured characteristic and zero otherwise. *Government security* is the interest rate on a Belgian government security with equal repayment duration as loan to firm, in basis points. *Term spread* is the yield on a Belgian government bond of 5 years—yield on treasury bill with a maturity of 3 months, in basis points. The number of observations is 15,044. We employ ordinary least squares estimation.

Variable Categories	Independent Variables	
<i>Relationship Variables</i>	<i>Main Bank</i>	−40.7*** (3.7)
	ln(1 + Duration of Relationship)	19.2*** (2.3)
<i>Loan Contract Characteristics</i>	<i>Collateral</i>	−50.9*** (8.3)
	ln(1 + Repayment Duration of Loan)	−92.5*** (9.3)
<i>Loan Purpose</i>	<i>Included Dummies</i>	<i>4 Loan Revisability***</i>
	Rollover	21.3*** (7.3)
<i>Firm Characteristics</i>	<i>Included Dummies</i>	<i>6 Loan Purpose***</i>
	Small Firm	44.0** (19.2)
	Medium Firm	−99.5*** (26.2)
	Large Firm	−170.2*** (51.4)
	Limited Partnership	−30.2 (18.7)
	Limited Partnership w/ES	−46.3* (24.7)
	Corporation	−116.2*** (21.1)
	Temporary arrangements	−35.1 (24.2)
	<i>Included Dummies</i>	<i>8 Postal Area***</i>
		<i>49 Industry***</i>
<i>Interest Rate Variables</i>	Government Security	0.5*** (0.1)
	Term Spread	0.4*** (0.1)
	<i>Included Dummies</i>	<i>2 Years***</i>
	<i>Intercept</i>	589.6*** (122.9)
	Adjusted R^2	0.222

*, **, and *** indicate significance at the 10%, 5%, and 1% level, two-tailed.

Economic message: Relationship banking has two sides: main-bank status can lower costs, but relationship duration can increase borrower lock-in. This distinction is central because location rents and information rents are both potentially present.

7.4 Table V: Borrowing Costs and the Role of Distance

Read:

- The coefficient on $\log(1 + \text{Distance to Lender})$ is negative in all models, ranging from about -4.3 to -14.2 basis points.
- The coefficient on $\log(1 + \text{Distance to Closest Competitors})$ is positive in all models, ranging from about 8.3 to 18.5 basis points.
- Main-bank borrowers continue to pay lower rates.
- Relationship duration continues to raise rates.
- The number of competitors is not significant in Model I.
- HHI is positive and significant in several models, but the economic magnitude is small.
- Adding postal-zone random effects, real-estate prices, and urban interactions leaves the distance results largely intact.

Detailed interpretation: Table V is the central empirical table. The opposite-signed distance coefficients are exactly the pattern predicted by spatial price discrimination. A borrower farther from the lending branch

pays less, while a borrower farther from competing banks pays more. This means that the lending bank uses the borrowers' local outside-option geography when setting the loan rate.

Economic message: The table converts geography into loan pricing. Borrowers close to the lender but far from competitors are locally captive, so they pay higher rates.

Table V
Borrowing Costs and the Role of Distance

The table lists the coefficients from regressions with the *loan rate* until the next revision, in basis points, as the dependent variable. *Distance to lender* is the shortest traveling time, in minutes. *Distance to closest competitors* is the shortest traveling time to the closest quartile competitor in the borrower's postal zone, in minutes. *Main bank* equals 1 if the bank considers itself as the main bank and zero otherwise. *Duration of relationship* is the length of relationship with the current lender, in years. *Number of competitors* is the number of branches (minus the lender's) in the borrower's postal zone. *Herfindahl-Hirschman Index* is the summed squares of bank market shares, by number of branches, in each postal zone. *Average real estate price* is recorded per postal zone in 1995, in millions of BEF. *Urban* equals 1 if the firm is located in an area with more than 250,000 inhabitants and zero otherwise. The number of observations is 15,044. We employ ordinary least squares estimation.

Independent Variables	I	II	III	IV	V	VI	VII
Distance							
ln(1 + Distance to Lender)	-4.3*	-5.4**	-10.3***	-5.4**	-8.3***	-14.2***	-12.8**
	(2.5)	(2.5)	(3.7)	(2.5)	(2.2)	(5.5)	(5.5)
ln(1 + Distance to Closest Competitors)	16.1***	16.6***	18.5***	16.7***	8.3***	14.2***	12.9**
	(3.8)	(3.6)	(4.0)	(3.6)	(2.2)	(5.5)	(5.5)
Relationship Variables							
Main bank	-69.9***	-41.1***	-53.0***	-41.1***	-43.9***	-44.4***	-44.9***
	(3.7)	(12.7)	(3.8)	(3.7)	(3.9)	(3.9)	(4.6)
ln(1 + Duration of Relationship)	18.8***	18.8***	23.9***	18.9***	18.6***	18.4***	18.7***
	(2.3)	(2.3)	(2.4)	(2.3)	(2.3)	(2.5)	(2.5)
Main bank × ln(1 + Distance to Lender)						11.1**	11.1**
						(4.6)	(4.6)
Main bank × ln(1 + Distance to Closest Competitors)						-11.1**	-11.1**
						(4.6)	(4.6)
ln(1 + Duration of Relationship) × ln(1 + Distance to Lender)						-0.1	-0.0
						(2.7)	(2.7)
ln(1 + Duration of Relationship) × ln(1 + Distance to Closest Competitors)						0.1	0.0

Table V: Borrowing Costs and the Role of Distance, part 1

Competition							
ln(1 + Number of Competitors)	-0.4						
	(2.6)						
Herfindahl-Hirschman Index	35.3***						
	(15.2)						
Postal zone random effects		Yes ^b					
Firm location							
Average real estate price ^a			-1.1				
			(2.2)				
Urban							14.6
							(24.8)
Urban × ln(1 + Distance to Lender)							-10.5
							(7.3)
Urban × ln(1 + Distance to Closest Competitors)							23.8
							(15.0)
Loan contract characteristics (including four loan revisability dummies), Loan purpose firm characteristics (including 8 postal area and 49 industry dummies), Interest rate variables (including 2-year dummies), and intercept	Yes	Yes	Yes ^c	Yes	Yes	Yes	Yes
Equality restriction(s), F-statistic							
Adjusted R ²	0.227	0.223	0.143 ^d	0.223	0.222	0.222	0.224

^a, **, and *** indicate significance at the 10%, 5%, and 1% level, two-tailed.
^bIn millions of BEF.
^cLagrange multiplier test of effects versus no effects = 390.1***, and Hausman (1978) test of fixed versus random effects = 35.0.
^dExcluding postal area and industry dummies.

Table V: Borrowing Costs and the Role of Distance, part 2

7.5 Table VI: Loan Size, Duration, and Collateral

Read:

- Including loan size does not eliminate the distance effects: distance to lender remains negative and distance to competitors remains positive.
- The distance effects are strongest among small loans and weaken as loan size rises.
- Short-duration loans show much stronger distance effects than longer-duration loans.
- Uncollateralized loans show strong spatial price discrimination; collateralized loans show much weaker and insignificant distance effects.
- Loan size is negatively related to loan rates across relevant specifications.

Detailed interpretation: Table VI is one of the most important mechanism tables. Fixed transportation costs should matter more for small loans than for large loans because a given trip cost is larger relative to loan size. The fact that distance coefficients decline with loan size and are strongest for short, uncollateralized loans supports the transportation-cost interpretation.

Economic message: The heterogeneity in Table VI tells us that distance is not just a proxy for local risk. It matters most where a borrower's practical cost of switching lenders is likely to be most important.

7.6 Table VII: Loan Type/Purpose, Firm Type, and Relative Distance

Read:

- Lines of credit show weaker or opposite distance patterns, while term loans show strong spatial price discrimination.
- Noncapital-expenditure loans show strong distance effects; capital-expenditure loans show much weaker effects.

Table VI
Loan Size, Duration, and Collateral

The table lists the coefficients from regressions with the *loan rate* until the next revision, in basis points, as the dependent variable. *Distance to lender* is the shortest traveling time, in minutes. *Distance to closest competitors* is the shortest traveling time to the closest quartile competitor in the borrower's postal zone, in minutes. *Main bank* equals one if the bank considers itself as the main bank and zero otherwise. *Duration of relationship* is the length of relationship with the current lender, in years. *Herfindahl-Hirschman Index* is the summed squares of bank market shares, by number of branches, in each postal zone. *Loan size* is in millions of BEF. We employ ordinary least squares estimation.

Independent Variables	Incl. Loan Size	By Loan Size (LS), in millions of BEF			By Duration of Loan (DL), in years			Collateral	
		LS ≤ 0.2	0.2 < LS ≤ 2	2 < LS	DL < 0.55	0.55 ≤ DL	No	Yes	
ln(1 + Distance to Lender)	-13.6*** (2.4)	-15.0*** (4.1)	-4.0* (2.1)	-0.7 (2.3)	-21.0*** (4.1)	-6.3** (2.8)	-12.9*** (2.9)	-2.3 (2.0)	
ln(1 + Distance to Closest Competitors)	13.6*** (2.4)	15.0*** (4.1)	4.0* (2.1)	0.7 (2.3)	21.0*** (4.1)	6.3** (2.8)	12.9*** (2.9)	2.3 (2.0)	
Main bank	-43.0*** (3.9)	-32.9*** (6.4)	8.3** (3.5)	-6.6* (4.0)	-53.1*** (6.5)	-35.3*** (4.5)	-45.6*** (4.7)	-7.6** (3.2)	
ln(1 + Duration of Relationship)	29.3*** (2.4)	26.0*** (4.4)	14.4*** (2.0)	10.7*** (2.1)	36.9*** (4.3)	24.7*** (2.6)	29.9*** (3.0)	1.8 (1.7)	
Herfindahl-Hirschman Index	10.3 (16.0)	32.0 (29.8)	14.0 (13.6)	38.3*** (14.3)	19.3 (27.4)	1.6 (17.9)	-11.0 (19.5)	37.0*** (12.9)	
Loan size	-22.8*** (1.1)				-56.8*** (2.6)	-11.4*** (1.1)	-140.6*** (3.5)	-4.7*** (0.5)	
Interest rate variables (including 2-year dummies) and intercept	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
Number of observations	15,044	5,850	7,344	1,850	6,698	8,346	11,073	3,971	
Equality restriction, F	0.115	3.268	4.616	1.717	0.491	0.411	0.091	1.733	
Adjusted R^2	0.084	0.011	0.136	0.065	0.089	0.085	0.175	0.447	

*, **, and *** indicate significance at the 10%, 5%, and 1% level, two-tailed.

Table VI: Loan Size, Duration, and Collateral

Table VII
Loan Type/Purpose, Firm Type, and Relative Distance

The table lists the coefficients from regressions with the *loan rate* until the next revision, in basis points, as the dependent variable. *Distance to lender* is the shortest traveling time, in minutes. *Distance to closest competitors* is the shortest traveling time to the closest quartile competitor in the borrower's postal zone, in minutes. *Main bank* equals one if the bank considers itself as the main bank and zero otherwise. *Duration of relationship* is the length of relationship with the current lender, in years. *Herfindahl-Hirschman Index* is the summed squares of bank market shares, by number of branches, in each postal zone. *Loan size* is in millions of BEF. We employ ordinary least squares estimation.

Independent Variables	Loan Type		Loan Purpose		Firm Type		Relative Distance	
	Lines of Credit	Term	Capital Expenditures	Noncapital Expenditures	SPB & SP *	Other	Lender = Closest	Lender ≠ Closest
ln(1 + Distance to Lender)	3.7 (2.5)	-10.5*** (2.7)	-0.2 (1.7)	-16.0*** (5.9)	-11.2*** (2.6)	-19.1*** (5.8)	-14.7*** (7.2)	-12.3*** (3.9)
ln(1 + Distance to Closest Competitors)	-3.7 (2.5)	10.5*** (2.7)	0.2 (1.7)	16.0*** (5.9)	11.2*** (2.6)	19.1*** (5.8)	14.7*** (7.2)	12.3*** (3.9)
Main bank	3.1 (4.0)	-34.3*** (4.4)	-5.0* (2.9)	-50.4*** (4.7)	-51.3*** (4.2)	-53.3*** (10.2)	-53.3*** (5.9)	-54.9*** (5.2)
ln(1 + Duration of Relationship)	9.3*** (3.5)	23.1*** (2.8)	3.9** (1.5)	33.4*** (3.0)	25.1*** (2.6)	24.4*** (6.5)	25.3*** (3.7)	31.8*** (3.1)
Herfindahl-Hirschman Index	22.6 (16.5)	-13.8 (18.0)	30.1** (11.3)	-5.9 (19.4)	10.0 (17.4)	-10.0 (38.2)	-5.2 (42.3)	11.2 (17.4)
Loan size	-5.8*** (0.6)	-224.5*** (4.1)	-37.0*** (0.4)	-73.3*** (2.5)	-11.0*** (1.8)	-38.6*** (1.5)	-11.0*** (2.2)	-18.6*** (1.3)
Interest rate variables (including 2-year dummies) and intercept	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	3,678	11,366	3,490	11,554	12,360	2,684	6,341	8,703
Equality restriction, F	0.173	1.798	3.453	1.861	0.411	0.999	0.064	0.039
Adjusted R^2	0.357	0.248	0.563	0.120	0.093	0.100	0.100	0.075

*, **, and *** indicate significance at the 10%, 5%, and 1% level, two-tailed.

*Single-person businesses operating as sole proprietorships.

Table VII: Loan Type/Purpose, Firm Type, and Relative Distance

- Sole proprietorships and other firm types both show distance effects, but magnitudes differ.
- Splitting by whether the lender is the closest branch yields similar distance signs, suggesting the result is not mechanically driven by choosing the closest bank.
- The equality restriction tests generally do not reject the idea that the lender and competitor distance coefficients are symmetric in magnitude.

Detailed interpretation: Table VII reinforces the mechanism. Spatial price discrimination is most visible in loan types where the borrowers' outside options and travel costs are salient. The result is weaker for products that involve deeper relationships, working-capital relationships, or capital expenditure financing, where transportation-cost based shopping may be less important.

Economic message: Banks price not only borrower risk but also borrower geography. The pattern varies by product because the relevance of local outside options varies by product.

8 Result map

- **Theory map:** Table I predicts which signs are consistent with transportation costs, monitoring costs, discriminatory pricing, and asymmetric information.
- **Setting:** Table II documents a dense branch network with many local competitors, making borrower-level geography meaningful.
- **Baseline data:** Table III shows that borrowers are near both lenders and competitors, and that loan contracts vary substantially.
- **Relationship pricing:** Table IV shows lower rates for main-bank borrowers but higher rates with relationship duration.
- **Main distance result:** Table V shows rates fall with distance to lender and rise with distance to competitors.
- **Mechanism:** Table VI shows stronger effects for small, short-term, and uncollateralized loans.
- **Product heterogeneity:** Table VII shows stronger effects for term and noncapital-expenditure loans than for lines of credit and capital-expenditure loans.
- **Interpretation:** The evidence is most consistent with transportation-cost based spatial price discrimination.

9 Short summary

- The paper studies whether local geography affects small-business loan pricing.
- It uses more than 15,000 loan contracts from a large Belgian bank and maps each borrower to the lender and nearby competitors.
- The main dependent variable is the loan rate until the next revision.
- The key explanatory variables are log travel time to the lending branch and log travel time to the closest quartile competitor.
- Loan rates decrease with distance to the lender and increase with distance to competitors.
- This sign pattern matches spatial price discrimination based on borrower transportation costs.
- Main-bank status reduces loan rates, but relationship duration increases them.
- The number of competitors alone is not a strong predictor; distance to competitors is more informative.
- Distance effects are strongest for transactional, short-term, uncollateralized, term, and noncapital-expenditure loans.
- Distance effects are weaker for broad relationships, collateralized loans, lines of credit, and capital expenditure loans.
- Robustness checks using alternative distance measures, location controls, and firm characteristics preserve the main result.
- The paper concludes that physical branch proximity remains important in the pricing of bank credit.

10 Conclusion

10.1 Core conclusion

Degryse and Ongena provide comprehensive evidence that geography affects the pricing of bank loans. Loan rates fall as borrowers are farther from the lending branch and rise as borrowers are farther from competing branches. This pattern is statistically robust and economically meaningful.

10.2 Economic intuition

A borrower located near the lending branch but far from competitors has weak outside options because visiting alternative lenders requires more travel time. A lender with local discretion can exploit this by charging a higher rate. A borrower located near competitors can credibly shop around, so the lending bank must offer a lower rate.

10.3 Incremental contribution revisited

The paper's incremental contribution is to document spatial price discrimination in financial contracting with contract-level data. It does not merely show that distance affects bank choice; it shows that distance affects the price of credit. It also separates location rents from relationship rents by showing that relationship duration and borrower geography have distinct effects on loan rates.

10.4 Why the evidence is persuasive

The evidence is persuasive because the two distance variables have opposite signs predicted by theory, the coefficients survive extensive controls, and the heterogeneity across loan size, duration, collateral, loan type,

and loan purpose supports a transportation-cost interpretation. The authors also benchmark the magnitudes in terms of travel costs and bank rents, making the estimates economically interpretable.

10.5 What the evidence does not prove

The paper cannot fully rule out all unobserved borrower-quality or branch-selection mechanisms. Distance is not randomly assigned, and borrower branch choice may reflect unobserved preferences or private information. The setting is one Belgian bank, so external validity must be assessed carefully. Nevertheless, the consistency across specifications and heterogeneity tests makes the spatial-price-discrimination interpretation compelling.

10.6 Policy and banking implications

The results imply that branch networks can discipline local credit-market power. Consolidation, branch closures, or the weakening of local alternatives may raise borrowing costs for small firms, especially for transactional and uncollateralized credit. Conversely, dense branch networks can improve credit-market competition even when the total number of banks appears unchanged.

10.7 Takeaway

The paper's lasting message is that distance is not dead in banking. Local physical proximity shapes small-firm loan rates because it changes borrower outside options and lender market power. Geography remains a source of bank rents even in a densely branched market.